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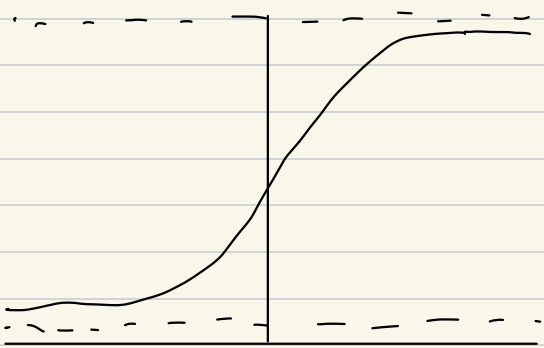
Logistic Regression

Linear Regression:

↳ $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots \beta_n x_n + \varepsilon$

Sigmoid Function:

↳ $y = \frac{1}{1 + e^{-x}}$



Logistic Regression Formula:

↳ $y = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots \beta_n x_n + \varepsilon)}}$

Logit Formula (Log Odds) Formula:

↳ $\ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 + \dots \beta_n x_n + \varepsilon$

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$$\ln\left(\frac{p}{1-p}\right) = z \rightarrow \text{Where } z \text{ equals linear component}$$

$$\frac{p}{1-p} = e^{z}$$

$$p = (1-p)e^{z}$$

$$p = e^{z} - p \cdot e^{z}$$

$$p + p \cdot e^{z} = e^{z}$$

$$p(1+e^{z}) = e^{z} \qquad p = \frac{e^{z}}{1+e^{z}}$$

$$p = \frac{1}{1+e^{-z}} \quad \underline{\underline{QED}}$$

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$\frac{1500}{90} = 16 \text{ games}$

135, 144

Columns to keep (Passer Model):

0 - 9, 14, 32, 53, 60

Use logistic scaling so all values are proportionate rather than evenly spread (when using linear)

Rating Formula $= \frac{1}{1 + e^{-k*z}} \times 100$

k = Gradient, z = Standardised Score

$$\frac{1}{1+e^{-x}}$$

$x = 0, \; r = 50$

$x = 1, \; r \approx 65$

$x = 2, \; r \approx 80$

Best Formula for Passing Rating (Per 90):

$$\frac{100}{1+e^{(-0.8 * x)}}$$

Passing Weights (14 rds):

$\frac{1}{14} \approx 0.07 \ldots$

6 equal, 8 less

$10x + 4y = 100$

$6x + 8y = 100$

84 16

0.14 0.02

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Next Steps

- Create rating models for all types of stats
- Create for each position to give more relevant rating
- Create regression models using SS rating as target variable
- Create regression (maybe Poisson) models for FPL (expected points)

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Goalkeeper Model

Passing Code:

3-9, 114, 115

Percentages Already Have:

Long Balls, Passes

Need:

Own Half, Opposition Half

Weights $4x + 5y = 100$ $(x > y)$

$x = 15 \quad y = 8$

$7x + 2y = 100$

$x = 12 \quad y = 8$

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$$5x + 2y + 2z = 100$$

$x = 14$, $y = 10$, $z = 5$

Final Weightings:

Final Third, Loose Balls, Opposition Half = 14

All Passes = 10 — 32

Own Half = 5

$$3x + 3y + 2z = 100$$

$x = 17$, $y = 9$, $z = 2.5$

$$68 + 27 + 5 = 100$$

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After dropping accurate passes and long balls

7 eqs: $2x + 3y + 2z = 100$ 50 (36, 14)

$x = 25, y = 12, z = 7$

New features (Few with small collinearity)

Accurate passes, Accurate pass %, Long ball %, Final third passes?

Index: 3, 5, 8, 9

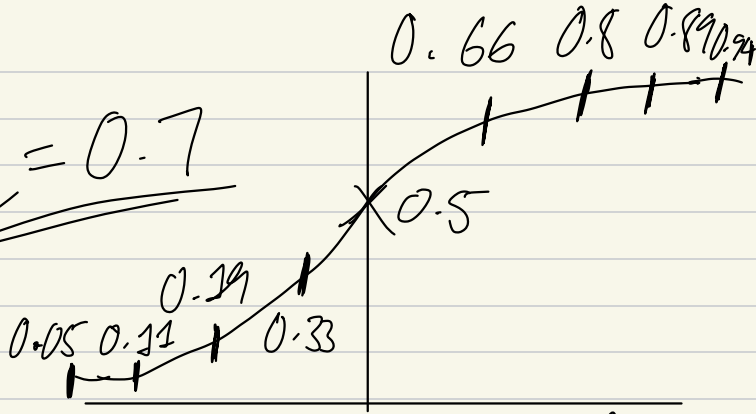
Weights: 10, 40, 10, 40

VIF all below 5 ✓

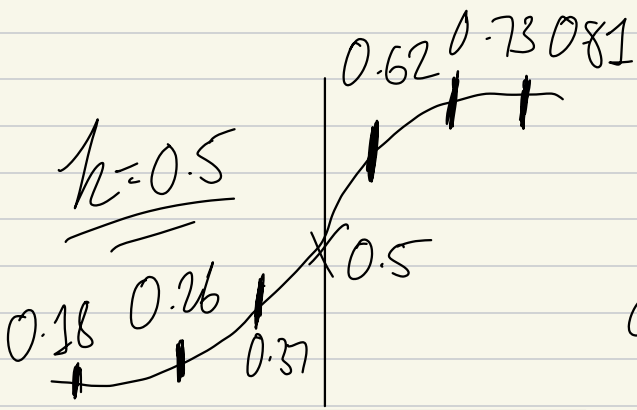
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$$\frac{1}{1+e^{-kx}}$$

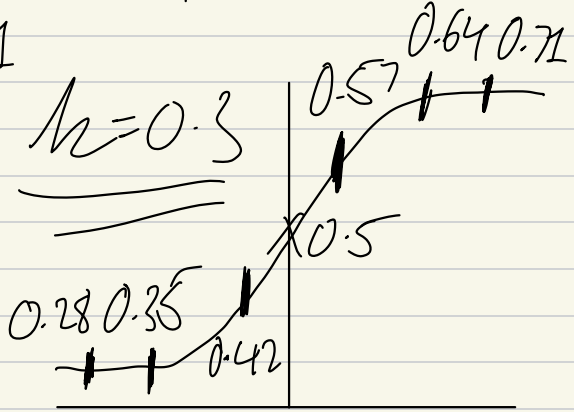
$k = 0.7$



$k = 0.5$



$k = 0.3$



$x > 0,\ k = 0.8$

$x < 0,\ k = 0.3$

$$f(x) \begin{cases} \dfrac{1}{1+e^{-0.8x}} & x \geq 0 \\ \dfrac{1}{1+e^{-0.3x}} & x < 0 \end{cases}$$

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Saves Rating

(5)

Features: Penalties Saved,
Penalty Save % (Penalty Saved / Faced),
Total Saves,
Save % (Saves / (Goals + Saves)),
Goals prevented

Sweeper Rating

(3)

Features: Succ. runs-out,
Succ. runs-out % (Succ / Total)
Recoveries

Aerial Rating

(3)

Features: Punches, High Claims,
High Claim %
↳ (High Claim / Not Claimed + Claimed)

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Saves Rating Weights

PS	PS%	S	S%	GP	Tot
0.05	0.05	0.1	0.35	0.45	1

Sweeper Rating Weights

SRO	SRO%	R	Tot
0.6	0	0.4	1

Aerial Rating Weights

P	HC	HC%	Tot
0.5	0.5	0	1

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Overall Rating Weights

Pass.	Sav.	Sweep.	Aerial	Tot
0.2	0.45	0.15	0.2	1

In future, increase k for when $x > 0$

(i.e. increase rating for those above average)